ClimateAdapt Pune: Evidence-Led Climate Action Report

Pilot Study: Pune Urban Core, Maharashtra | Generated: 18 October 2026 17:13 UTC

Executive Summary: This report translates real municipal drainage statistics (OpenCity Pune), OpenStreetMap infrastructure assets (5,779 trees, 267 medical facilities, waterways), Open-Meteo 7-day atmospheric forecasts, and Google Gemini AI into actionable climate resilience priorities for Pune's electoral wards. All findings distinguish measured infrastructure facts from model proxies, with automated MILP budget optimization and expert LLM strategic synthesis.

1. High-Priority Wards (Screening Indicators)

Scores are normalized 0-100 screening proxies. Higher values denote greater exposure or infrastructure deficit.

Ward / Prabhag	Heat Proxy	Flood Proxy	Water Stress	Green Deficit	Composite
Vadgaon Sheri - Ganesh N	58.4	75.2	52.0	95.0	68.8
Arunkumar Vaidya Stadium	64.4	54.8	60.0	95.0	65.0
Ghorpade Garden	63.8	54.6	60.0	93.3	64.4
Upper Indiranagar	58.4	59.1	60.0	95.0	64.4
Mohammadwadi	58.4	60.7	52.0	95.0	63.7
Vimannagar Sanjay Park	58.4	55.4	60.0	95.0	63.1

2. Prioritized Adaptation Action Plan (Budget Allocation)

Target Portfolio Budget: INR 5,000,000 | **Committed Allocation:** INR 5,000,000 (100.0% efficiency)

Intervention	Category	Units	Cost (INR)	Dept	Lead Time
Shaded Bus Stops & Public Co	Heat & Urban Co	12 station	■2,160,000	PMPML & PMC Health	2 mo
Cool Roof Reflective Coating	Heat & Urban Co	15 building	■1,800,000	PMC Building & Dev	1 mo
Decentralized Rainwater Harv	Water Security	2 building	■440,000	PMC Water Supply D	2 mo
Critical Stormwater Drain De	Flood & Stormwa	1 major dr	■350,000	PMC Drainage & Sew	1 mo
Urban Shade Corridors & Nati	Heat & Urban Co	1 km corri	■250,000	PMC Garden Departm	3 mo

3. Climate Stress Scenario Comparisons

Scenario	Heat Index	Flood Index	Water Stress	Composite Index
Current Baseline	58.4	75.2	52.0	68.8
Extreme Heatwave (+3.5°C)	78.4	75.2	67.0	78.0
Extreme Monsoon Cloudburst	48.4	100.0	42.0	76.0
Post-Adaptation Plan	39.6	75.2	49.8	61.6

4. Responsible Use & Data Quality Disclosures

• ***Distinction of Evidence:*** *Drainage statistics are verified 2015-2016 OpenCity PMC releases. Atmospheric values represent Open-Meteo multi-variable NWP forecast models and must not be cited as permanent climate normals.*
• ***Explicit Missing Data:*** *Local observation wells (GSDA telemetry) and uncalibrated DEM hydrological runoff grids remain unconfirmed and are strictly excluded from automated physical simulation to prevent false certainty.*
• ***Human Review:*** *Action allocations are exploratory decision-support calculations and require municipal site surveys before civil construction.*

5. AI Ward Resilience Brief: Ward #19 (Vadgaon Sheri - Ganesh Nagar)

Analysis Engine: Google models/gemini-3.8-flash | **Zone:** Nagar Road | **Evaluation Date:** 2026-10-18

1. What's Life Like in This Ward?

In Vadgaon Sheri - Ganesh Nagar, summer afternoons feel like an oven because there are practically zero trees or parks for shade across the entire area. When it rains, nearly every street sees instant *pani bharaav* because a shocking 98.2% of our roads have no roadside drains at all. For families living under tin roofs or in crowded slum pockets, the heat stays trapped inside all day, and overflowing dirty water during rains makes daily life miserable.

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2. Problems and What We Can Do About Them

Problem	How Bad Is It?	Why Does This Happen Here?	What Can Be Done?
Street Flooding	Terrible; only 1.8% of roads have working drains.	Rainwater has nowhere to escape, so even brief showers turn streets into mini-rivers.	Clean out the main blocked *naala* and fit iron trash screens to stop choke-ups.
Extreme Heat	Very high; temperatures reach 34.3°C with tin roofs baking inside.	Densely packed homes absorb all the sun's heat without any natural relief.	Coat building rooftops with white heat-reflecting paint and set up cool resting kiosks.
No Trees or Shade	Critical; officially 0 trees and 0 public parks are recorded.	Rapid construction has paved over every open mud patch and corner.	Plant a 1 km green pathway with sturdy local shade trees.
Dirty Water Overflow	High risk; 80% of slum homes lack underground sewer lines.	Toilet and kitchen waste often mix with open rainwater on roads.	Clean open drains urgently and screen out floating garbage.
Water Shortage	Looming problem; huge population of 32,318 with fast-falling groundwater.	Rain pours down and runs away as waste instead of sinking into the soil.	Set up rooftop rainwater collection tanks on municipal buildings.

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3. How Will the Money, Workers and Time Be Used?

Work To Be Done	How Many?	Money (Rs.)	Workers Needed	Time Needed	Who Is Responsible?
Shaded bus stops & cooling kiosks with cold drinking water	12 kiosks	Rs. 2,160,000	30 person-months	2 months	PMPML & PMC Health Department
White heat-reflecting paint for building rooftops	15 building clusters	Rs. 1,800,000	30 person-months	1 month	PMC Building & Development Department
Rooftop rainwater collection tanks on public buildings	2 installations	Rs. 440,000	6 person-months	2 months	PMC Water Supply Department
Cleaning major *naala* and installing trash screens	1 major drain section	Rs. 350,000	5 person-months	1 month	PMC Drainage & Sewerage Department
Tree planting along walking paths	1 km pathway	Rs. 250,000	4 person-months	3 months	PMC Garden Department / Tree Authority

- Total money being used:** Rs. 5,000,000 out of Rs. 5,000,000
- Money left over:** Rs. 0
- Workers needed:** 75.0 person-months
- Everything done in:** 3 months

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4. What Should the Ward Officer Do? (Step by Step)

1. **Week 1–2:** Clear garbage and silt from the main ward *naala*, install metal screens so trash stops blocking the water flow, and identify the 15 hottest tin-roof clusters.
2. **Month 1:** Send painter teams to finish the white heat-reflecting paint on the 15 building clusters, and start digging pits along the 1 km path for native trees.
3. **Month 2:** Build and open the 12 shaded cooling kiosks with cold drinking water near busy bus stops, and begin fixing rooftop rainwater tanks at municipal offices.
4. **Month 3:** Complete the rainwater collection tanks, finish planting all tree saplings with protective tree guards, and inspect the main drain before heavy rains begin.

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5. How Will Residents Know Things Are Getting Better?

- ■ After rain, streets should drain within 30-45 minutes instead of staying flooded for hours.
- ■ Homes with white-painted roofs should feel cooler in the afternoon, making rooms bearable without running fans at top speed.
- ■ Auto drivers, street vendors, and walkers will have cool, shaded kiosks with clean, cold drinking water to take a break from the harsh sun.